Supplementary Material

Systematic benchmarking of functional module partitioning
in GO pathway subnetworks

Supplementary Tables S1--S6

Supplementary Figures S1--S2

Total evaluation records: 3100 (main) + 528 (synthetic) + 317 (downstream)

Table S1. Cohort characteristics and pathway coverage

Accession	Disease	Median pathway genes	Samples	Pathways	Algorithm evaluations
GSE10072	Lung adenocarcinoma	91	107	7	7
GSE137342	Severe sepsis	124	57	6	6
GSE15653	Type 2 diabetes / obesity	117	18	6	6
GSE17536	Colorectal cancer	125	177	7	7
GSE19188	Non-small cell lung cancer	137	156	7	7
GSE21942	Multiple sclerosis	150	29	6	6
GSE25066	Breast cancer	115	486	7	7
GSE30153	Influenza infection	112	26	6	6
GSE36961	Hypertrophic cardiomyopathy	172	145	7	7
GSE38642	Type 2 diabetes (islets)	121	63	6	6
GSE39088	Rheumatoid arthritis	218	142	7	7
GSE39582	Colorectal cancer	102	560	6	6
GSE42568	Breast cancer	86	121	6	6
GSE55098	Obesity / metabolic	152	22	6	6
GSE61635	Systemic lupus erythematosus	31	129	4	4
GSE6764	Hepatocellular carcinoma	101	72	6	6

Table S2. Complete algorithm performance metrics

Algorithm	Modules	Modularity Q	Coherence z	Stability	Robustness	Antagonistic	n
CPM γ=0.20	140.440	0.029	3.769	0.904	0.874	0.994	100
CPM γ=0.10	111.830	0.093	2.325	0.873	0.919	0.968	100
Clique Percolation	46.500	0.046	2.225	0.915	0.934	0.295	100
CPM γ=0.05	85.920	0.151	1.755	0.867	0.951	0.874	100
K-means	5.620	0.140	1.397	0.739	0.874	0.613	100
Ward	5.620	0.141	1.391	0.663	0.836	0.589	100
OPTICS	12.360	0.145	1.130	0.286	0.630	0.821	100
Gaussian Mixture	5.620	0.140	1.128	0.654	0.772	0.549	100
DBSCAN	3.930	0.087	1.046	0.527	0.858	0.254	100
HDBSCAN	4.300	0.177	0.998	0.393	0.736	0.430	100
Walktrap	9.010	0.256	0.909	0.590	0.802	0.536	100
Leiden γ=1.0	4.610	0.290	0.906	0.659	0.885	0.541	100
Louvain γ=0.5	16.880	0.239	0.889	0.500	0.851	0.617	100
Louvain	4.680	0.290	0.866	0.644	0.891	0.547	100
Infomap	5.660	0.237	0.863	0.743	0.942	0.394	100
Fast Greedy	4.510	0.285	0.851	0.655	0.910	0.538	100
Label Propagation	2.680	0.221	0.831	0.677	0.911	0.098	100
Multilevel	4.630	0.289	0.820	0.646	0.841	0.590	100
Leiden γ=10.0	92.300	0.040	0.802	0.663	0.891	0.981	100
Leiden γ=2.0	26.910	0.200	0.796	0.646	0.875	0.793	100
Louvain γ=2.0	26.210	0.201	0.789	0.628	0.855	0.821	100
Affinity Propagation	19.430	0.221	0.727	0.683	0.820	0.847	100
Spectral	5.620	0.202	0.710	0.598	0.865	0.550	100
Leiden γ=0.5	2.250	0.242	0.683	0.681	0.894	0.169	100
NMF	5.590	0.183	0.667	0.501	0.790	0.681	100
Leading Eigenvector	3.990	0.266	0.659	0.596	0.909	0.509	100
Leiden γ=5.0	68.690	0.086	0.657	0.663	0.893	0.923	100
Mean Shift	6.200	0.018	0.426	0.549	0.899	0.109	100
WGCNA-style Hierarchical	3.000	0.003	0.033	0.613	0.892	0.000	100
Leiden γ=0.1	1.110	0.127	0.031	0.929	0.969	0.009	100
Spin Glass	1.000	—	-0.086	1.000	1.000	0.000	100

Table S3. Paired statistical tests

Metric	Pairs	Median (exp)	Median (ref)	p	q (FDR)	Cliff δ	Sig.
模块数	100	4.500	3.000	1.95e-16	4.87e-16	+0.870	yes
模块度Q	100	0.291	0.000	3.90e-18	1.95e-17	+1.000	yes
功能内聚z	100	0.689	0.045	2.74e-07	4.57e-07	+0.418	yes
稳定性	100	0.646	0.626	6.79e-02	6.79e-02	+0.133	no
拮抗拆分率	44	0.500	0.000	5.59e-07	6.98e-07	+0.727	yes

Experimental: Leiden γ=1.0. Reference: WGCNA-style hierarchical. Wilcoxon signed-rank, Benjamini-Hochberg correction.

Table S4. Synthetic benchmark results (ground truth available)

Algorithm	LFR mean ARI	LFR max ARI	Nested sub-group ARI	Modules
CPM γ=0.10	0.379	1.000	1.000	12.0
Infomap	0.000	0.000	1.000	12.0
Label Propagation	0.237	1.000	1.000	12.0
Leiden γ=2.0	0.603	0.992	1.000	12.0
Leiden γ=5.0	0.122	0.220	1.000	12.0
Leiden γ=1.0	0.927	1.000	0.943	11.3
Louvain	0.941	1.000	0.943	11.3
Walktrap	0.774	1.000	0.943	11.3
WGCNA-style Hierarchical	0.929	1.000	0.908	11.0
CPM γ=0.05	0.000	0.000	0.863	10.3
Leiden γ=0.5	0.000	0.000	0.602	6.0
Spectral	0.876	1.000	0.602	6.0
Ward	0.724	1.000	0.551	6.0
K-means	0.641	1.000	0.377	6.0
Gaussian Mixture	0.598	1.000	0.352	6.0

LFR: block model across fuzziness $\mu \in \{0.10 – 0.70\}$. Nested: recovery of 12 ground-truth sub-groups (the decisive benchmark).

Table S5. Null-model comparison

Cohort	Pathway	Observed Q	Null A (rewired)	Null B (permuted)	z (A)	z (B)
GSE137342	apoptotic process (GO:0006915)	0.173	0.170	0.419	0.62	-12.42
GSE137342	cellular response to oxidative stress (G	0.165	0.215	0.510	-7.21	-11.96
GSE137342	chemokine-mediated signaling pathway (GO	0.315	0.544	0.583	-4.39	-5.59
GSE137342	inflammatory response (GO:0006954)	0.295	0.226	0.456	10.09	-6.25
GSE137342	regulation of I-kappaB kinase/NF-kappaB	0.171	0.147	0.417	6.52	-10.18
GSE137342	regulation of immune response (GO:005077	0.304	0.280	0.460	1.95	-6.56
GSE137342	response to cytokine (GO:0034097)	0.165	0.205	0.470	-6.22	-13.20
GSE25066	apoptotic process (GO:0006915)	0.366	0.785	0.482	-12.74	-3.94
GSE25066	cellular response to oxidative stress (G	0.213	0.704	0.537	-6.31	-8.46
GSE25066	chemokine-mediated signaling pathway (GO	0.344	0.648	0.566	-3.32	-5.02
GSE25066	inflammatory response (GO:0006954)	0.294	0.644	0.432	-15.51	-6.82
GSE25066	regulation of I-kappaB kinase/NF-kappaB	0.268	0.727	0.485	-14.55	-8.37
GSE25066	regulation of immune response (GO:005077	0.308	0.501	0.474	-7.44	-6.39
GSE25066	response to cytokine (GO:0034097)	0.485	0.717	0.517	-7.48	-0.90
GSE39582	apoptotic process (GO:0006915)	0.544	0.706	0.436	-6.03	4.70
GSE39582	cellular response to oxidative stress (G	0.500	0.730	0.510	-5.86	-0.34
GSE39582	chemokine-mediated signaling pathway (GO	0.529	0.621	0.561	-1.44	-0.71
GSE39582	inflammatory response (GO:0006954)	0.182	0.453	0.437	-15.07	-8.41
GSE39582	regulation of I-kappaB kinase/NF-kappaB	0.490	0.637	0.436	-7.21	2.59
GSE39582	regulation of immune response (GO:005077	0.263	0.394	0.474	-6.26	-7.27
GSE39582	response to cytokine (GO:0034097)	0.483	0.674	0.476	-6.80	0.26
GSE6764	apoptotic process (GO:0006915)	0.286	0.448	0.428	-8.59	-7.51
GSE6764	cellular response to oxidative stress (G	0.324	0.542	0.477	-6.70	-6.44
GSE6764	chemokine-mediated signaling pathway (GO	0.470	0.624	0.558	-2.96	-1.60
GSE6764	regulation of I-kappaB kinase/NF-kappaB	0.389	0.474	0.419	-5.51	-1.66
GSE6764	regulation of immune response (GO:005077	0.404	0.439	0.479	-1.72	-3.00
GSE6764	regulation of inflammatory response (GO:	0.335	0.459	0.459	-11.54	-4.71
GSE6764	response to cytokine (GO:0034097)	0.412	0.522	0.466	-3.45	-2.43

These z-scores are computed at the 75th-percentile edge-weight threshold and are highly threshold-dependent (see Table S6): the sign of the comparison reverses across the 50th-95th percentile range. Modularity comparisons against null models should therefore not be reported as a robust property of the data.

Table S6. Modularity is unstable with respect to graph construction

Edge-weight threshold	Q (weighted)	Modules	Q (uniform weights)	Modules	Weight contribution
50th	0.269	5.5	0.097	4.5	64%
75th	0.297	14.0	0.184	12.6	38%
85th	0.329	23.4	0.246	22.8	25%
90th	0.362	30.1	0.292	29.5	19%
95th	0.419	40.9	0.366	40.9	12%

Mean Q rises from 0.269 to 0.419 (+56%) across the threshold range, while the comparison against a degree-preserving null model reverses sign. Replacing all edge weights with a constant reduces mean Q from 0.335 to 0.237, i.e. weights contribute 29% of modularity independently of topology.

Supplementary Figures

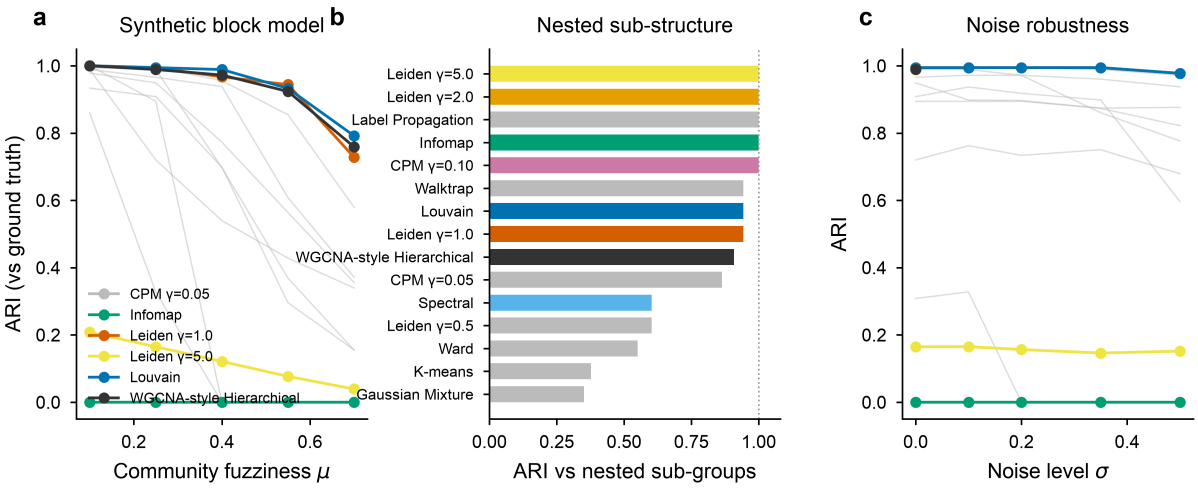


Figure S1. Synthetic benchmark accuracy across fuzziness and noise levels.

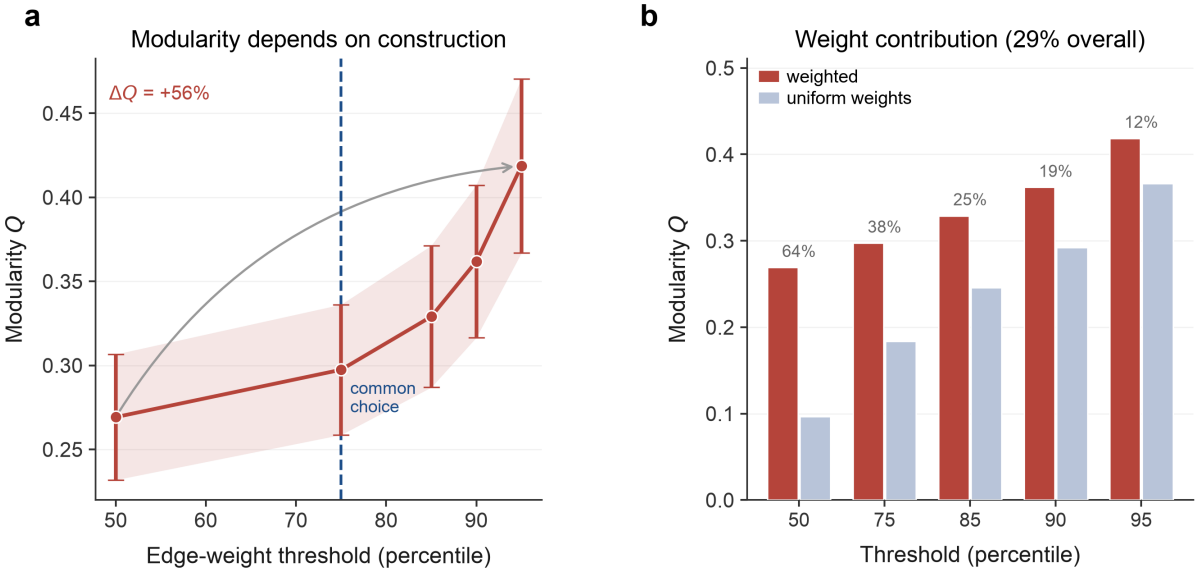


Figure S2. Modularity is unstable with respect to graph construction. Left: mean modularity rises monotonically with the edge-weight threshold. Right: weighted versus uniform-weight modularity, showing the 29% contribution of weights independently of topology.